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MILITARY HANDBOOK

AIRCRAFT FASTENERS INSTALLATION AND REMOVAL METHODS FOR MS14218 120° RIVETS

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DEPARTMENT OF DEFENSE
WASHINGTON 25 D.C. 20361

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Aircraft Fasteners Installation and Removal Methods for MS14218 120° Rivets

1. This standardization handbook was developed by the Department of Defense in accordance with established procedure.

2. The publication was approved on 15 August 1984 for printing and inclusion in the military standardization handbook series.

3. This handbook provides installation and removal methods for 120° rivets. It will provide valuable information and guidance to aircraft mechanics at Navy, Army and Air Force Rework Facilities.

   This handbook is not intended to be referenced in procurement specifications except for information purposes, nor shall it supersede any specification requirements.

4. Every effort has been made to include the latest information on installation and removal methods of 120° rivets. Mechanics should review this handbook periodically. Users of this handbook are encouraged to report any errors or recommendations to: Commanding Officer, Naval Air Engineering Center, Code 93, ESSD, Lakehurst, NJ 08733.

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1. SCOPE

1.1 Scope. This handbook covers the installation and removal methods for 120° (MS14218) rivets.

2. REFERENCED DOCUMENTS

MILITARY

STANDARDS

MS14218 Rivet, Solid, 120° Flush Interference Shear Head

SPECIFICATIONS

MIL-P-23377 Primer Coating, Epoxy Polymide, Chemical and Solvent Resistant

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3. Rivet application and description. The rivets are for use in structures where flushness, fatigue life, static joint strength, and corrosion resistance are of primary importance. The major features are a 120° flush, shear low-profile head with a pronounced land around the head diameter and a "ring dome" head surface. The "ring dome" flattens during riveting, expanding the head diameter into interference with the hole counterbore (see Figures 1 and 2).

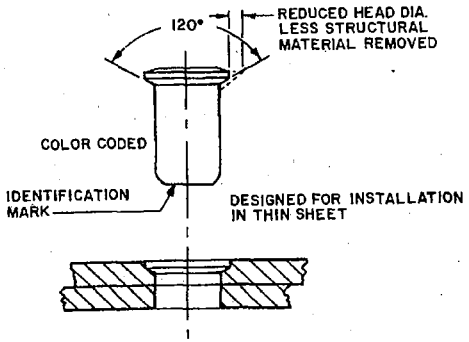


FIGURE 1. MS14218 rivet and hole.

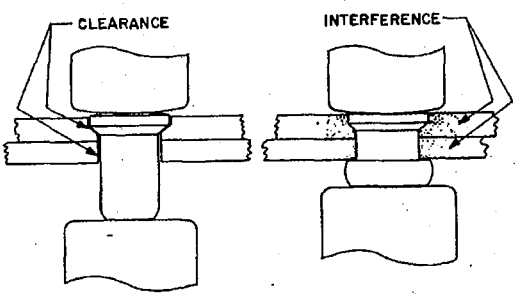


FIGURE 2. Rivet interference fit.

3.1 Tools. The recommended power tools for drilling holes for 120° (MS14218) rivets are:

1. Air or electric drill motor.
2. Straight drill of appropriate diameter and length.
*3. FZA-20-(style) micro-stop cage.
*4. FZP-(size) 120 degree piloted cutter.
*5. FZDG-(size) dual gage.

*See drawings on pages 20-22.

Any suitable air or electric powered drill motor may be used to drill holes. The motor speed (max RPM) for counterboring and countersinking in different materials is specified in 4.1.2.2.1.

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4 Rivet installation.

4.1 Hole preparation. Before installation, the rivet hole and counterbore shall be coated with primer in accordance with MIL-P-23377 or equivalent.

4.1.1 Hand drilling. Holes, countersinks and counterbores shall be drilled as specified in Figure 3 and Table I.

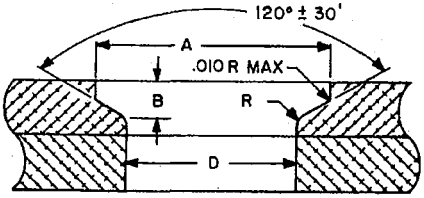


FIGURE 3. Recommended hole preparation.

TABLE I. Hole dimensions.

DIA DASH NO.	NOM DIA	A		B	D		STD DRILL NO.	R* RAD
		MIN	MAX	±.002	MIN	MAX		±.004
-3	3/32	.1235	.1265	.022	.098	.104	40	.008
-4	1/8	.1625	.1655	.027	.128	.135	30	.008
-5	5/32	.2105	.2140	.035	.161	.167	20	.009
-6	3/16	.2585	.2625	.044	.193	.200	10	.012
-7	7/32	.3125	.3170	.053	.228	.235	1	.015
-8	1/8	.3585	.3640	.061	.257	.265	F	.018
-9	9/32	.4055	.4110	.069	.290	.298	L	.020
-10	5/16	.4545	.4600	.077	.316	.327	O	.022

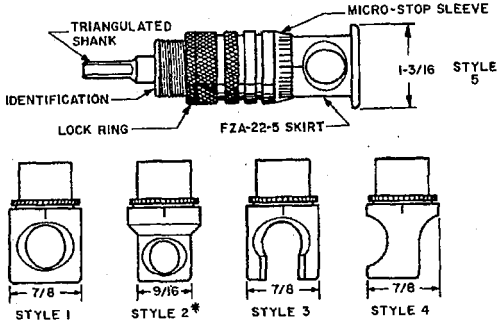
*"R" radius segment varies with "D" dia size.

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4.1.2 FZA-20 micro-stop cage. The FZA-20 micro-stop cage incorporates a precision adjustment to prevent counterboring too deep.

Skirt styles may be exchanged. When the micro-stop is adjusted to the same micro-line setting, the hole dimensions will be within 0.001-inch.

The FZA-20 cage must be used with a FZP piloted cutter to ensure consistent quality of countersinks.



OPTIONAL SKIRT STYLES

*Use Style 2 skirt in confined spaces and concave areas.

FIGURE 4. Micro-stop cage.

4.1.2.1 Set-up instructions. Loosen the lock ring. Pull back the spring-loaded micro-sleeve and tighten selected skirt firmly against the stop. Back off 3 or 4 micro-lines from the fully tightened position. Release the spring-loaded micro-sleeve to interlock the skirt. Tighten the lock ring.

This will be approximately the correct setting to machine a counterbore to the depth specified in Table I.

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4.1.2.2 FZP piloted cutter. The FZP piloted cutter is designed to work with the FZA micro-stop cage assembly to produce uniform holes for the installation of 120° (MS14218) rivets in aluminum alloy and titanium alloy structures.

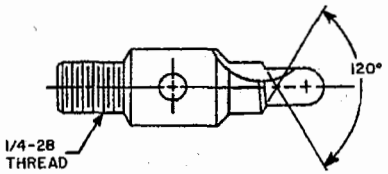


FIGURE 5. Piloted cutter.

Thread the FZP piloted cutter into the FZA micro-stop cage until it is firmly seated. (Take special care to make sure there is no foreign material on the mating surfaces of the micro-stop cage spindle and the piloted cutter. Foreign material may cause cutter wobble or counterbore depth variation.)

Any diameter FZP cutter may be exchanged in a properly adjusted FZA cage without affecting the set depth by more than 0.002-inch. However, good shop practice calls for a test installation after any tooling change.

See 4.1.2.4 for drilling instructions.

4.1.2.2.1 Cutter size and speed. Cutter size and speed shall be in accordance with Table II.

TABLE II. Cutter size and speed.

NOMINAL HOLE DIAMETER	CUTTER FZP-SIZE	RECOMMENDED MAX RPM FOR ALUMINUM	RECOMMENDED MAX RPM FOR TITANIUM
3/32	-3		
1/8	-4	3,000	650
5/32	-5		
3/16	-6		
7/32	-7	2,500	500
1/4	-8		
9/32	-9		
5/16	-10	2,000	400

CAUTION: Do not exceed recommended RPM (revolutions per minute). Exceeding maximum recommended RPM will result in reduced cutter life and oversize counterbores.

DO NOT USE: Modified cutters or micro-stop cages; dull, damaged or wobbling cutters; washers or spacers under the cutters; or high speed drill motors when cutting titanium.

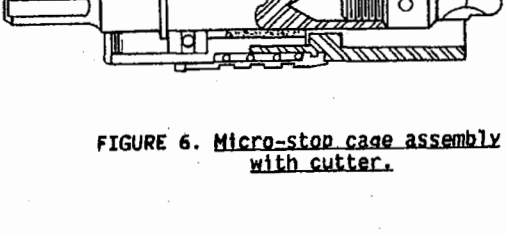


FIGURE 6. Micro-stop cage assembly with cutter.

4.1.2.3 Inspection procedures using a FZDG dual gage.

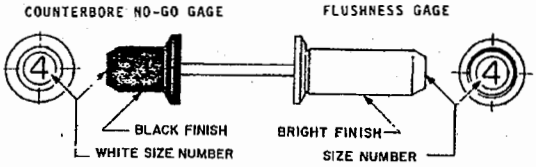


FIGURE 7. FZDG dual gage.

4.1.2.3.1 Counterbore diameter inspection. Check for oversize counterbore diameter by inserting black end of the gage into drilled hole (see Figure 8). If the head of the black gage enters the counterbore, the hole is too large (not within tolerance of Table I).

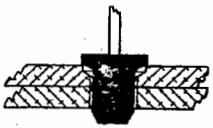


FIGURE 8. Counterbore inspection.

4.1.2.3.2 Flushness inspection. Check for flushness by inserting the long, bright finished end of the dual gage into the drilled hole (see Figure 9). A gage flushness within a tolerance of +.001 -.003 indicates that the installed rivet will conform to the flushness requirements of Figure 12.

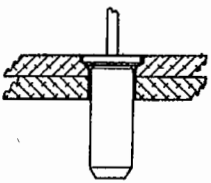


FIGURE 9. Flushness inspection of straight surface.

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To inspect the flushness on a curved surface, measure at the lowest point of the skin surface at the edge of the counterbore (see Figure 10).

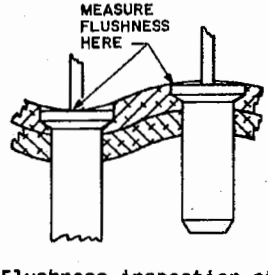


FIGURE 10. Flushness inspection of curved surface.

4.1.2.4 Hand drilling set-up and inspection procedures. Thread the FZP piloted cutter into the FZA micro-stop cage. Attach the cage/cutter assembly to the drill motor. Conduct initial drilling and countersink operation on a test plate. (When possible, use a coolant on the cutter to extend cutter life.)

Check counterbore diameter with the short, black end of the FZDG dual gage as indicated in 4.1.2.3.1 and Figure 8.

Check the counterbore depth with the long, bright finished end of the FZDG dual gage as indicated in 4.1.2.3.2 and Figure 9.

If a FZDG dual gage is unavailable, use a MS14218 rivet (see Figure 11 and Table III). The P dimension should be maintained to hold rivet flushness within the specified tolerance after riveting (see Figure 12).

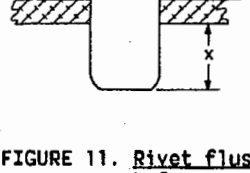


FIGURE 11. Rivet flushness before installation.

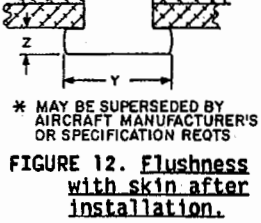


FIGURE 12. Flushness with skin after installation.

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TABLE III. Hand riveted dimensions.

NOM DIA	P ±.002	X +.0625 -.0313	Y MIN	Z MIN
3/32	.003	.094	.132	.033
1/8	.004	.125	.175	.045
5/32	.004	.156	.219	.057
3/16	.005	.188	.264	.070
7/32	.007	.222	.307	.083
1/4	.008	.250	.350	.095
9/32	.009	.283	.394	.107
5/16	.009	.314	.438	.125

4.1.3 Hand riveting instructions. Install the correct rivet size into the structure. Upset the rivet using the proper size rivet gun and bucking bar to ensure upsetting the rivet in less than three seconds (see Figure 13).

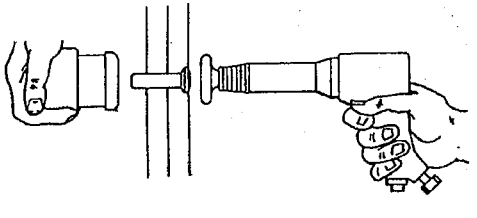


FIGURE 13. Rivet installation.

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a. Select a FLAT or BELL rivet set approximately twice the diameter of the rivet head.

b. A bell set is recommended when riveting in skin quality areas.

c. Select a bucking bar weight approximately one-half the rivet gun weight.

d. Ensure that the working surfaces of rivet sets and bucking bars are polished and free of defects.

e. The rivet head is expanding simultaneously while upsetting the bucktail.

f. Allow the rivet gun to begin before exerting bucking force on the terminal end of the rivet shank. This ensures that the rivet head will be fully seated.

g. Rivet head tipping can be minimized and driving time reduced when the center of the rivet set is aligned with the center of the rivet.

h. Inspect the upset button for conformance to Y and Z dimensions (see Figure 12 and Table III).

i. Inspect the flushness of the installed rivet head (see Figure 12 and Table III). Additional minor adjustments may be required due to surface variation.

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4.1.4 Riveting in sealant area. When riveting in wet sealant areas, remove excess sealant from the skin surface under the structure and from the rivet before driving the rivet. Do not use solvents that might remove or damage the skin's protective finish. Keep rivet sets and bucking bars free from contamination.

It is critical to seat the rivet head and squeeze out excess sealant before riveting. This ensures proper head seating and flushness of the installed rivet.

4.1.5 Drivmatic or squeeze riveting. Drivmatic installation must meet the same hole dimensions specified in Table I.

Seat the rivet head in the countersink before upset. The upper ram surface that contacts the rivet must be in the same plane or slightly below the plane of the upper pressure foot that contacts the structure. Unless otherwise specified in engineering drawings, the formed button dimensions (Y values) shall be in accordance with Table IV.

It is critical that hole dimensions, and Y and Z dimensions are in accordance with Tables I and IV during riveting with automatic or squeeze riveting equipment. Flushness requirements (P) shall be in accordance with Figures 14 and 15 and Table IV.

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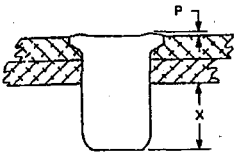


FIGURE 14.
Rivet flushness before installation.

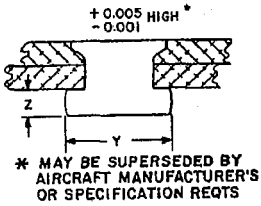


FIGURE 15.
Flushness with skin after installation

TABLE IV. Squeeze rivet dimensions.

NOM DIA	P ± 0.002	X +.0625 -.0313	Y MIN	Z MIN
3/32	.003	.094	.141	.033
1/8	.004	.125	.188	.045
5/32	.004	.156	.234	.057
3/16	.005	.188	.282	.070
7/32	.007	.222	.329	.083
1/4	.008	.250	.375	.095
9/32	.009	.283	.422	.107
5/16	.009	.314	.465	.125

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4.2 120° (MS14218) rivet removal. When removing rivets, it is important to avoid damaging the original hole or adjacent structure. The following methods are recommended for drilling out countersunk MS14218 rivets:

Centerpunch the center of the manufactured head and drill the punch location through the head of the rivet but not into the rivet shank.

CAUTION: Do not drill below head depth into shank. If the structure is light, back up the rivet with a bucking bar when centerpunching. Always use a drill of the same size as the original hole.

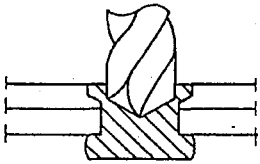


FIGURE 16. Drill out.

Break off the head by inserting a pin punch in the drilled portion of the rivet. While holding the punch perpendicular to the structure, punch out the rivet. Support the opposite side of the structure to avoid dimpling the structure. Pry out the remaining head carefully so as not to damage or elongate the counterbore.

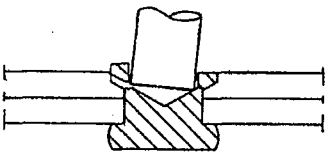


FIGURE 17. Break out.

In heavy material, it may be necessary to drill into the rivet shank before driving it out. Use a drill bit 1/32" under the rivet size.

IMPORTANT: Recheck holes for compliance with Table I to ensure dimensions are within tolerance before reinstalling rivets.

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4.3 Rivet material substitution. Aluminum alloy substitution is permitted in accordance with the aircraft manufacturer's specification or Table V.

TABLE V. Aluminum alloy substitution.

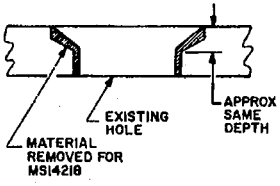
EXISTING RIVET ALUMINUM ALLOY	SUBSTITUTE RIVET ALUMINUM ALLOY
2024-T4 (DD)	7050-T73 (E)
2017-T4 (D)	7050-T73 (E)
2219-T81 (A)	7050-T73 (E)

4.4 Rivet retrofit and repair. When possible, use the same rivet diameter and length that was originally installed. Rivet diameter substitution is permitted only when the hole dimensions of Table I cannot be maintained. Oversize replacement shall be as follows:

Retrofit of NAS1097 with MS14218 allows one dash number diameter increase without significant depth change in accordance with Table VI.

TABLE VI. NAS1097 replacement table.

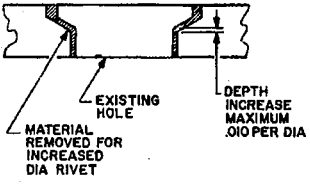
EXISTING NAS1097	MS14218 REPLACEMENT
-3	-4
-4	-5
-5	-6
-6	-7
-7	-8



Replace existing MS14218 with one dash number diameter increase in accordance with Table VII.

TABLE VII. MS14218 oversize replacement.

EXISTING NAS1097	REPLACEMENT MS14218
-3	-4
-4	-5
-5	-6
-6	-7
-7	-8
-8	-9
-9	-10



When countersink depth cannot exceed the original depth, "shaving" the head to the required flushness tolerance is permitted (see Figure 18)

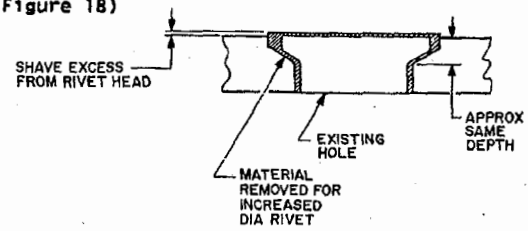


FIGURE 18. Shaving rivet head.

4.5 Rivet cross-reference. The rivets listed in Table VIII are of the same configuration and are interchangeable when of the same size and material.

TABLE VIII. Rivet cross-reference.

MILITARY STANDARD	MS14218
BRILES RIVET CORP.	BRFZ Rev M
VOI-SHAN DIV. OF VSI CORP.	VRFZ Rev NC
BOEING AIRCRAFT CORP.	BACR15FV Rev B
McDONNELL DOUGLAS CORP.	3M1022 Rev A
NORTHROP CORP.	71R210 Rev 2

4.6 Tooling procurement. The tools depicted in this handbook are available from:

Briles Rivet Corporation
2640 Vista Pacific Drive
Oceanside, CA 92056-3599
Telephone: (619) 941-6500

VOI-SHAN Division of VSI Corp.
8463 Higuera Street
Culver City, CA 90230
Telephone: (213) 838-2131

Preparing Activity
Navy-AS
Project No. 5320-0149-01

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4.7 MS14218.

FED. SUP CLASS 5320

1. APPLICATION

1.1 THESE RIVETS ARE INTENDED FOR USE IN RIVETED STRUCTURES WHERE FLUIDNESS, FATIGUE LIFE, STATIC JOINT STRENGTH AND CORROSION RESISTANCE ARE OF PRIMARY DESIGN IMPORTANCE

1.2 RIVETS SHALL BE USED IN ACCORDANCE WITH MIL-STD-1515, REQUIREMENT 206

(A) 2. MILITARY INSTALLATION MANUAL MIL-HDBK-271-102

3. MATERIALS

3.1 ALUMINUM ALLOYS PER QQ-A-430	MATERIAL LETTER CODE	
3.1.1 2017-T4	D	
3.1.2 2117-T4	AD	
3.1.3 5056-H32	B	
3.1.4 7050-T73	E	
3.1.5 TITANIUM COLUMBIUM, 45 cb PER AMS 4982	T	

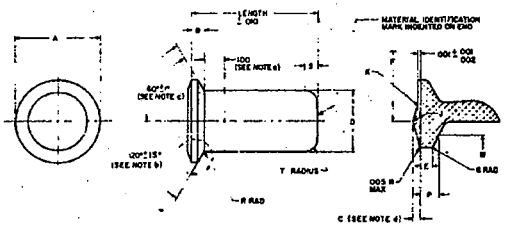
3.2 COATINGS, ALUMINUM ALLOYS

(A) 3.2.1 ANODIZED PER MIL-A-8625, TYPE IC, COLOR (SEE 5.1).

(A) 3.2.2 CHEMICAL SURFACE TREATMENT PER MIL-C-5541, CLASS 1A

4. HEAT TREATMENT IN ACCORDANCE WITH MIL-H-6088 FOR ALUMINUM ALLOYS. IN ACCORDANCE WITH AMS 4982 FOR TITANIUM COLUMBIUM ALLOYS.

5. ILLUSTRATION



(a) .001 SHANK DIAMETER INCREASE PERMISSIBLE WITHIN .100 INCH OF HEAD

(b) THE 120° CONICAL SURFACE OF THE HEAD SHALL BE ANNULAR WITHIN .003 AND CONCENTRIC WITH THE SHANK OF THE RIVET WITHIN .003 TOTAL INDICATOR READING

(c) HEAD COCKING ANGLE RELATIVE TO AXIS OF RIVET 1° MAXIMUM

(d) THE RIVET CROWN (C DIMENSION) SHALL JOIN THE RIVET HEAD FLAT SURFACE SMOOTHLY WITHOUT ABRUPT SHOULDERS

(A) DENOTES CHANGES

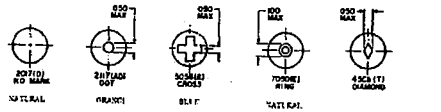
APPROVED 4 AUGUST 1982 REVISED (A) 15 MAY 84

P.C. NAVY - AS Other Cust ARMY - AV AIR FORCE - 11	TITLE RIVET, SOLID 120° FLUSH INTERFERENCE SHEAR HEAD	MILITARY STANDARD MS14218
PROCUREMENT SPECIFICATION MIL-R-5674	SUPERSEDES	SHEET 1 OF 3

Source: https
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PROJ. MAT CLASS 5320

5.1 MATERIAL IDENTIFICATION SYMBOLS. SYMBOLS INDENTED .030 MAXIMUM DEPTH.



6. TABLE I. RIVET DIMENSIONS

DIA DASH NO.	NOM DIA	A +.000 -.002	B (REF)	C	D +.002 -.001	E MIN	F ±.010	G RAD MAX.	J RAD ±.010	K RAD MIN	F MAX	F MIN	R RAD ±.004	S ±.010	T RAD ±.010	V GAGE DIA MAX	V GAGE DIA MIN
-3	3/32	.123	.022	.004 .008	.094	.012	.094	.005	.016	.051	.0188	.0167	.008	.023	.029	.1102	.1100
-4	1/8	.162	.027	.004 .008	.125	.014	.125	.005	.016	.047	.0223	.0202	.011	.031	.039	.1462	.1460
-5	5/32	.210	.035	.005 .009	.156	.017	.156	.005	.020	.040	.0270	.0248	.014	.039	.049	.1892	.1890
-6	3/16	.258	.044	.005 .009	.187	.021	.188	.006	.025	.078	.0289	.0266	.018	.047	.059	.2422	.2420
-7	7/32	.312	.053	.006 .010	.222	.024	.219	.007	.025	.094	.0418	.0387	.022	.054	.069	.2671	.2669
-8	1/4	.358	.061	.006 .011	.250	.027	.250	.009	.031	.125	.0470	.0444	.026	.062	.078	.3038	.3036
-9	9/32	.405	.069	.007 .012	.283	.030	.281	.010	.040	.125	.0538	.0511	.029	.068	.090	.3315	.3313
-10	5/16	.454	.077	.007 .012	.310	.033	.313	.010	.040	.125	.0465	.0435	.031	.078	.098	.4245	.4243

7. TABLE II. RIVET MECHANICAL PROPERTIES

ALLOY AND TEMPER	TENSILE STRENGTH psi MIN	YIELD STRENGTH psi MIN	ELONGATION IN 2 INCHES OR 4D % MIN	UNDRIVEN SHEAR STRENGTH psi MIN	UNDRIVEN SHEAR STRENGTH psi MAX
5056-H32	44,000	------	---	24,000	------
2117-T4	38,000	18,000	18	26,000	------
2017-T4	55,000	32,000	12	33,000	39,000
2050-T73	68,000	58,000	10	41,000	48,000
45 Cb	85,000	60,000	15	50,000	56,000

8. TABLE III. LENGTH NUMBERS

LENGTH (a)	DIAMETER DASH NUMBERS -3	-4	-5	-6	-7	-8	-9	-10
.125	2	-	-	-	-	-	-	-
.156	2R	-	-	-	-	-	-	-
.188	3	3	3	-	-	-	-	-
.219	3R	3R	3R	-	-	-	-	-
.250	4	4	4	4	-	-	-	-
.281	4R	4R	4R	4R	-	-	-	-
.312	5	5	5	5	5	-	-	-
.343	5R	5R	5R	5R	5R	5R	-	-
.375	6	6	6	6	6	6	6	-
.406	6R	6R	6R	6R	6R	6R	6R	-
.438	7	7	7	7	7	7	7	7
.470	7R	7R	7R	7R	7R	7R	7R	7R
.500	8	8	8	8	8	8	8	8
.531	8R	8R	8R	8R	8R	8R	8R	8R

APPROVED 4 AUGUST 1961 REVISED A

CUSTODIANS NAVY - AS Other Coast ARMY - AV AIR FORCE - 11	TITLE: RIVET, SOLID, 120° FLUSH INTERFERENCE SHEAR HEAD	MILITARY STANDARD MS14218
PROCUREMENT SPECIFICATION MIL-R-1474	SUPERSEDES	SHEET 2 OF 3

FED. SUP CLASS 5320

LENGTH (a)	DIAMETER DASH NUMBERS -3	-4	-5	-6	-7	-8	-9	-10
.562	9	9	9	9	9	9	9	9
.593	9R	9R	9R	9R	9R	9R	9R	9R
.625	10	10	10	10	10	10	10	10
.687	11	11	11	11	11	11	11	11
.750	12	12	12	12	12	12	12	12
.812	-	13	13	13	13	13	13	13
.875	-	14	14	14	14	14	14	14
.937	-	15	15	15	15	15	15	15
1.000	-	16	16	16	16	16	16	16
1.062	-	-	17	17	17	17	17	17
1.125	-	-	18	18	18	18	18	18
1.187	-	-	19	19	19	19	19	19
1.250	-	-	20	20	20	20	20	20
1.312	-	-	21	21	21	21	21	21
1.375	-	-	22	22	22	22	22	22
1.437	-	-	23	23	23	23	23	23
1.500	-	-	24	24	24	24	24	24
1.625	-	-	-	26	26	26	26	26
1.750	-	-	-	28	28	28	28	28
1.875	-	-	-	-	-	30	30	30
2.000	-	-	-	-	-	32	32	32

(a) RIVET BODY SHALL HAVE A RADIUS END EXCEPT THAT LENGTH IN SHADED AREA MAY HAVE A SQUARED END.

(A) REVISED ALL LENGTHS TO 0.0625 INCREMENTS; R DESIGNATES 0.031 INCREASED LENGTHS.

9. EXAMPLE OF PART NUMBER.

MS14218 AD 10 - 12 R

- R — INCREASED LENGTH BY .031-INCH (A)
- 12 — LENGTH DASH NUMBERS IN .0625-INCH INCREMENTS (A)
- - — "-" ANODIZED; "C" CHEMICAL SURFACE TREATMENT (A)
- 10 — DIAMETER DASH NUMBERS IN .031-INCH INCREMENTS
- AD — MATERIAL LETTER CODE
- MS14218 — MILITARY STANDARD NUMBER

10. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

11. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS OR REQUEST FOR PROPOSAL EXCEPT THAT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.

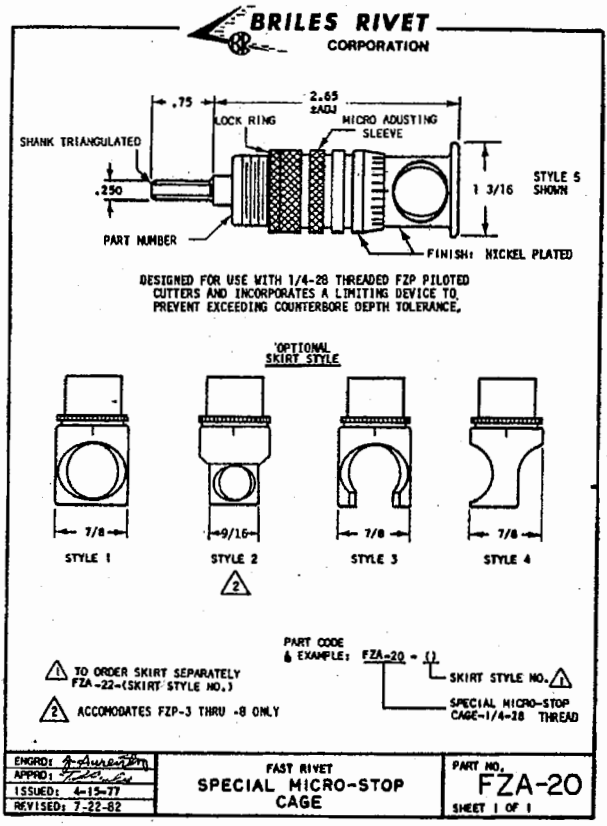
12. THE RIVETS ARE A PROPRIETARY PRODUCT OF BRILES RIVET CORPORATION AND ARE COVERED BY PATENT NO. 4,157,668 EXPIRING 3 JULY 1946. THE UNITED STATES GOVERNMENT DOES NOT HAVE A ROYALTY-FREE LICENSE.

APPROVED 4 AUGUST 1982 REVISED A

P.A. NAVY - AS Other Cog: ARMY - AV AIR FORCE - 11	TITLE: RIVET, SOLID, 120° FLUSH INTERFERENCE SHEAR HEAD	MILITARY STANDARD MS14218
PROCUREMENT SPECIFICATION: MIL-R-5674	SUPERSEDES:	SHEET 3 OF 3

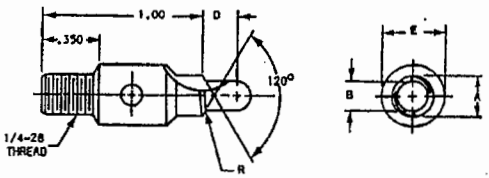
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4.8 FZA-20 micro-stop cage.



4.9 FZP 120° piloted cutter.

BRILES RIVET CORPORATION



DIA DASH NO.	NOM DIA	*A	B	D	E	R RAD
-3	3/32	.124	.097	.094	.375	.008
-4	1/8	.163	.127	.125	.375	.008
-5	5/32	.211	.160	.156	.375	.009
-6	3/16	.259	.192	.188	.375	.012
-7	7/32	.313	.227	.219	.375	.015
-8	1/4	.359	.256	.250	.410	.018
-9	9/32	.406	.289	.281	.50	.020
-10	5/16	.455	.315	.313	.50	.022

*COUNTERBORE 'A' DIA 100% TESTED TO INSURE PRODUCTION COUNTERBORES WITHIN TOLERANCE AS SPECIFIED IN FZH-101 DESIGNED FOR CUTTING ALUMINUM ALLOY.

PART CODE & EXAMPLE: FZP -6

-6 — NOMINAL DIA IN 1/32"

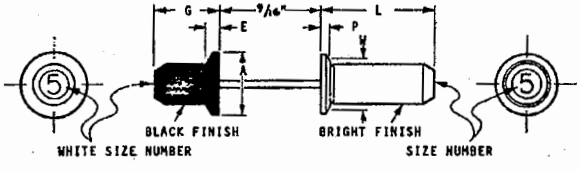
FZP — 120° PILOTED CUTTER. 1/4-28 THREAD SHEAR HEAD

FZP PRECISION CUTTERS ARE DESIGNED TO BE USED ONLY IN THE SPECIAL MICRO-STOP CAGE FZA-20.

ENGRD:	FAST RIVET	PART NO.
APPRD:	120° PILOTED CUTTER	FZP
ISSUED: 4-1-78		SHEET 1 OF 1
REVISED: 8-15-82 "J"		

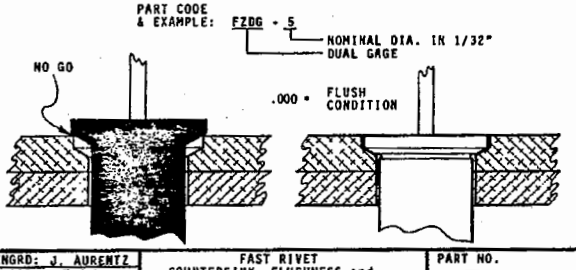
4.10 FZDG dual gage.

BRILES RIVET CORPORATION



DUAL GAGE

DIAMETER DASH NUMBER	NOMINAL DIAMETER	A NO GO	E	G	W	P	L
-3	3/32	.1265	.020	.25	.1101	.0198	.50
-4	1/8	.1655	.025	.25	.1461	.0233	.50
-5	5/32	.2140	.033	.25	.1891	.0279	.50
-6	3/16	.2625	.042	.25	.2421	.0298	.50
-7	7/32	.3170	.051	.28	.2670	.0424	.56
-8	1/4	.3640	.059	.31	.3037	.0482	.62
-9	9/32	.4110	.067	.34	.3314	.0525	.68
-10	5/16	.4600	.075	.38	.4244	.0480	.75



ENGRD: J. AUREMTZ APPRD: F.G. ISSUED: 12/08/82 REVISED: NC	FAST RIVET COUNTERSINK FLUSHNESS and COUNTERBORE MAXIMUM DIAMETER DUAL GAGE	PART NO. FZDG SHEET 1 of 1

Source: h
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